

Smart Metro Ticket Booking System

PROJECT DESIGN PHASE

Date	
Team ID	
Project Name	Smart Metro Ticket Booking System
Maximum Marks	4 Marks

2. Proposed Solution

2.1 Solution Overview

The Smart Metro Ticket Booking System is a digital, self-service ticketing solution built on the ServiceNow platform. It allows commuters to select their source and destination stations, travel date, and number of passengers through a guided Service Catalog form. The system automatically calculates the fare using Flow Designer, generates a unique digital ticket with an embedded QR code, and instantly delivers the ticket and booking confirmation through the Notification module. This removes the need to visit a physical counter entirely and gives every commuter a fast, accurate, and accessible booking experience.

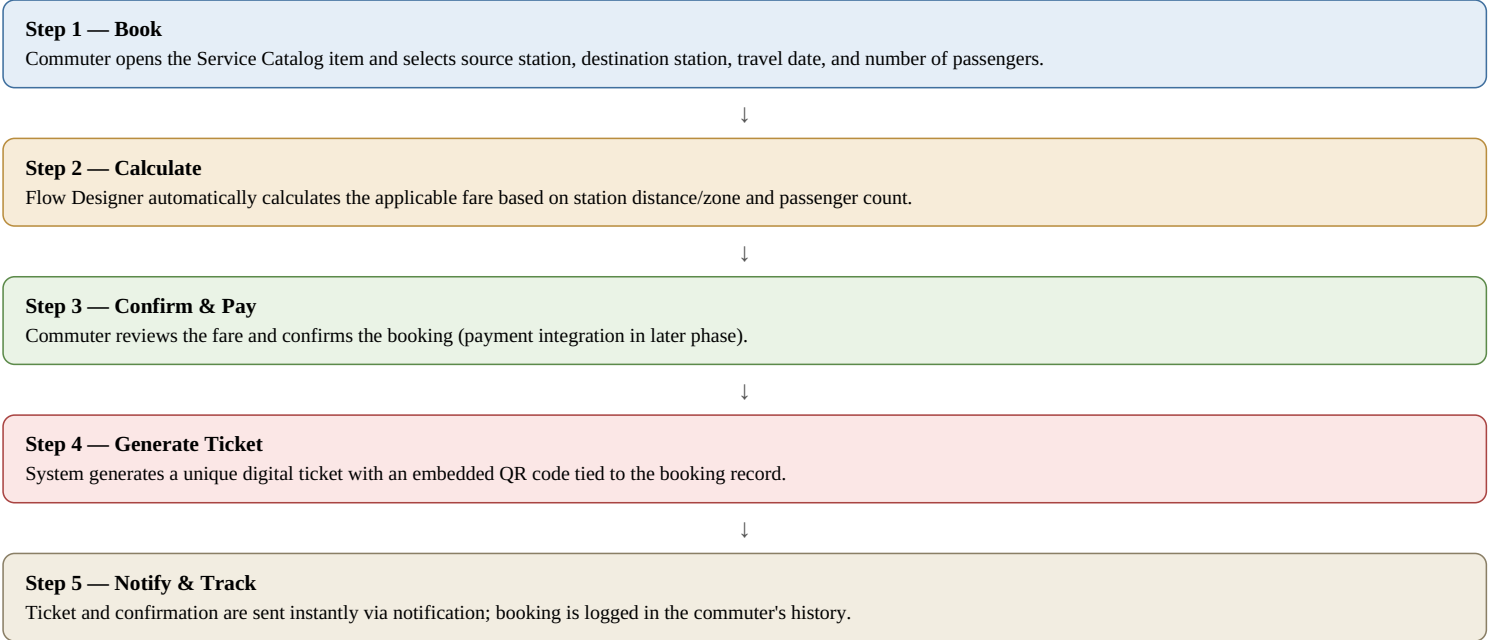
2.2 Solution Statement

"A ServiceNow-powered self-service platform that lets any commuter book a metro ticket in under a minute, with automatic fare calculation and an instant QR-code digital ticket — anytime, from anywhere, without visiting a counter."

2.3 Core Features (MVP Scope)

Feature	Description	Priority
Service Catalog booking form	Guided form to select source station, destination station, travel date, and number of passengers	P0
Automated fare calculation	Flow Designer workflow computes fare based on station distance/zone and passenger count	P0
QR-code digital ticket generation	Unique digital ticket with an embedded QR code generated automatically on successful booking	P0
Instant notifications	SMS/Email/Push notification delivers the digital ticket and booking confirmation immediately	P0
Booking history tracking	Every booking is tracked as a Service Catalog request record, viewable by the commuter	P0
Payment gateway integration	UPI/card/wallet integration for instant online payment during booking	P1
Cancellation and refund workflow	Self-service cancellation with automated refund processing	P1
Admin dashboard	Dashboard for metro staff to monitor bookings, revenue, and station-wise traffic	P1

2.4 How the Solution Works



2.5 Unique Value Proposition

- **Speed:** A metro ticket can be booked in under a minute, from anywhere, before reaching the station.
- **Accuracy:** Automated fare calculation removes manual, counter-based fare errors.
- **Accessibility:** A single, consistent digital flow works equally well for daily commuters, first-time riders, senior citizens, and differently-abled passengers.
- **No physical dependency:** A QR-code digital ticket replaces tokens and paper tickets that can be lost or damaged.
- **Platform efficiency:** Built entirely on native ServiceNow components, keeping the solution lightweight and maintainable.

2.6 Expected Outcomes

- Reduced queue time and counter congestion at metro stations, especially during peak hours.
- Fewer fare-related errors and disputes due to automated calculation.
- Lower operational cost for the metro authority in maintaining counters, staff, and token inventory.
- A more inclusive travel experience for elderly and differently-abled commuters.
- Higher adoption of metro rail as a preferred mode of daily transport.